From calibrated image to inspectable candidates

Two signal paths converge on the same matcher; mapping combines results after fitting each window.

1 Inputs and calibration

DM4 metadata or imported raster pixels with manual scale-bar calibration. Real-space units select FFT analysis; reciprocal units select direct diffraction.

Folder catalogues, locally generated previews and optional demo downloads feed image selection. Original pixels remain the measurement source.

2 Browser and local service

Browser sends source, ROI and explicit settings. A threaded local HTTP service validates requests and manages project state, jobs and exports.

Display contrast, palette and marker motion are separate from numerical processing. Stale interactive replies cannot overwrite a newly selected image.

3 Extract measured profiles → panel 2

Real space: crop → mean removal → optional Hann → FFT → radial background subtraction → smoothing → threshold components.

Direct diffraction: calibrated detector coordinates → masked multiscale spot detection → optional shared-centre refinement → manual peak edits.

4 Generate, fit and score → panels 3–5

Editable material cells and enabled HCP/BCC/FCC/primitive ORT templates. Fit in-plane rotation and allowed metric changes; assign profiles one-to-one.

Return positional agreement, scale costs, weak/missing evidence and screens. Geometry-only grids are a separate diagnostic, not a prepartition of the crystal evidence.

5 Map independent windows → panels 6–7

Uniform/adaptive regions or overlapping square windows on square/hexagonal centres. Spawned workers run the same numerical pipeline.

Batch runs compile per-image and by-sample summaries. Overlap maps average support/ranks, not FFT arrays. PCA/NMF is proposed, not implemented.

6 Inspect, reproduce and retain → panel 8

Raw/processed views, measured and predicted markers, vector annotations, per-fit tables and hover previews expose the evidence.

Project entries retain source/ROI/settings; JSON records numerical results and provenance. PNG/GIF exports are presentations. Candidate maps are not validated phase fractions.

Signal preparation and observed profiles

Coordinates are reciprocal cycles per nm (no 2π); opposite spots count as one profile.

1 FFT branch

F = FFT{(I − mean I) Hx Hy}; power = |F|²; L = log(1 + power). Mean removal and Hann taper are optional.

Sampling: Δgx = 1/(Nx sx), Δgy = 1/(Ny sy). The detector subtracts radial mean(L), then Gaussian-smooths by σ bins. Raw and processed displays remain separately available.

2 Direct-detector branch

qx = (x − cx)sx; qy = (cy − y)sy. No FFT is applied. Centre and reciprocal calibration are common to all candidate phases.

Masked Gaussian differences at multiple spot widths are checked against local response and detector-intensity contrast. Smoothing scales can run in threads; crystal fitting is sequential within this request.

3 FFT peak admission

Within gmin ≤ |q| ≤ gmax, cutoff T = max(floor, fraction × strongest processed signal). Connected regions above T contribute one maximum.

Minimum strength, sub-bin position refinement, minimum separation and the profile-count cap follow. Profile weight = clip(enrichment − 2, 1, 6). Estimated width and raw power are retained.

4 Direct peaks, edits and centre

Accepted spots are localized and deduplicated. Weight = clip(log(1 + local contrast), 1, 6). Only observed spots are drawn.

Manual additions/removals and masks are saved. Added observations bypass automatic contrast checks. Opposite-spot midpoints may refine one shared centre; competing centres retain the starting estimate.

5 Resolution and matching tolerance

FFT width estimate: 1.44 Δg with Hann, 0.886 Δg without; Δg uses the larger axis interval. Direct mode estimates spot width.

Matching tolerance τ = max(0.10 nm^{-1}, estimated width). It is distinct from detector separation and manual snap/removal radii. Enlarging a crop narrows FFT bins; it does not guarantee a better local phase assignment.

6 Output and limits

Profiles contain qx, qy, g, d = 1/g, weight, radius/intensity descriptors and provenance. Friedel-related observations are grouped rather than counted twice.

Fitting limits and masks govern assessable positions. Beam-stop-hidden predictions are not invented measured peaks. Thresholding, smoothing and the finite band can exclude real evidence.

Crystal predictions and metric fitting

Finite zone templates predict allowed positions; they do not simulate diffraction or HRTEM intensity.

1 Reciprocal vectors and zones

Cubic: q = (h,k,l)/a. HCP internal coordinates: q = ((2h+k)/(√3 a), k/a, l/c). ORT: q = (h/a,k/b,l/c).

Integer reflections obey the internal zone law hu + kv + lw = 0. Project the zone plane into two dimensions. HCP labels are converted to four-index notation for display.

2 Reflection filters

BCC: h+k+l even. FCC: h,k,l all even or all odd. HCP ideal basis: exclude odd l when h−k is divisible by 3.

Generic primitive orthorhombic: no basis-specific absences. A rectangular projection alone does not establish a unique 3D cell or identify a chemical phase. Search contains a finite, unequal library of zones.

3 Rotation and scale search

Fixed scale, shared automatic scale, independent HCP a*/c*, or overall + HCP c/a. ORT can fit independent visible cell dimensions.

Ordinary reciprocal bounds ×0.85–1.15; Unknown ×0.5–2 with overall cost disabled. Predictions move; measured reciprocal coordinates do not. Unobservable axes are explicitly not measurements.

4 Seeds and positional refinement

Seed angles/scales from a bounded set of measured profiles, then bounded weighted positional refinement and reassignment.

Legacy seeds are supplemented by up to two orientations with >2° angular difference and ≤80% support overlap. This improves exploration without providing exhaustive orientation search or guaranteed weak-pattern recovery.

5 Observable HCP metric

For sa and sc reciprocal multipliers, a = a0/sa and c = c0/sc. In overall + c/a mode: s = √(sa sc), r = sa/sc = (c/a)/(c0/a0).

The zone-plane metric is recomputed for anisotropic changes. For basal-zone data insensitive to c, c/a remains unconstrained and receives no ratio cost.

6 Assignment and retention → panel 4

Distance dij = min(|oi − pj|, |oi + pj|). Quality is Gaussian within τ and zero outside; dummy assignment slots permit unmatched observations.

Each profile and predicted node is used at most once per fit. Scale/evidence costs rank explored positional solutions after refinement; they are not the continuous refinement objective.

Scores, screens and automatic strength

Rank is an adjustable heuristic, not a calibrated likelihood or a phase probability.

1 Detected evidence and position

W = Σwi; C = Σmatched wi / W. Q = Σmatched wi exp[−½(dij/(τ/2))²] / W.

An angular chance allowance D depends on predicted-node density at each observed radius. Position P = Q − D. RMS = √mean(dij²), unweighted across matched profiles.

2 Scale preferences

Overall + c/a: S = λ[ln(s)/ln(1.05)]² + μ[ln(r)/ln(1.05)]². Cubic fits and c-insensitive HCP omit the ratio term.

Legacy independent-axis mode uses a mean squared fractional axis deviation normalized to 5%. These preferences are not physical strain limits; zero weights remove costs, not search bounds.

3 Weak and missing predictions

At assessable unmatched predictions, weak maxima are assigned locally and compared with 11 rotated controls. They contribute a bounded bonus B; they do not satisfy the strong-profile screen.

Unsupported positions cost M = Σunsupported 2^[−(g/rhalf)²] / Nassessable. rhalf = 0 restores uniform cost. This radial falloff is heuristic, not contrast-transfer correction.

4 Rank and acceptance

R = P − S + ηB − ηM. Strong screen: ≥4 independent matches, directional spread >0.5, C >0.30 and P >0.20. Final rank must also exceed 0.20.

Assessability excludes masked, out-of-band or unresolved prediction sites. Missing spots are not crystallographic extinction evidence. Weak maxima cannot rescue a fit with too few strong profiles.

5 Auto · dominant rank

Prepare admitted peak pool once. Try distinct minimum-strength cutoffs; match crystal candidates at each. Prefer a passing result, then highest top rank, then more profiles.

The denominator changes as peaks are removed. A higher winning rank can coincide with loss of a weaker second pattern. This mode deliberately selects the dominant interpretation.

6 Auto · geometry (1–2 patterns)

Search cutoff subsets using freely fitted grid evidence normalized to the full admitted pool. Count shared evidence once; subtract chance and per-grid costs.

A screened second grid contributes only when it improves the objective. Run crystal matching once on the selected subset. Neither Auto mode jointly optimizes multiple crystal patterns.

Separate interpretations from separate patterns

Crystal candidates use the observed set directly; geometry grids diagnose periodicity independently.

1 Crystal ranking

Retain and rank screened crystal fits. Primary is the highest-ranked accepted model. Rejected candidates remain inspectable.

A candidate explaining one part of a mixed pattern still uses whole-set coverage. This can penalize a physically real weak component. No geometry A/B membership constrains the crystal assignment.

2 Separate secondary evidence

A secondary crystal fit must itself pass and add ≥3 unique profiles, nonparallel spread >0.5 and >15% of total observed weight beyond the primary.

Different phase → two candidate structures. Same phase → multiple orientations. The method reports at most a primary/secondary interpretation; candidate alternatives are not an arbitrary-component mixture model.

3 Competing explanations

Different-phase accepted rivals close in rank and sharing substantial primary evidence can produce an ambiguous result rather than two structures.

A further shared-evidence check retains ambiguity when the leader matches exactly the same profiles but uses more scale parameters or more unsupported predictions than an accepted rival.

4 Geometry-only grids

Fit two freely chosen reciprocal basis vectors and their integer combinations. Basis seeds, density limits and positional screens suppress trivial over-dense explanations.

The geometry score includes a 0.03 per-grid cost and separate coverage criteria. Primary and secondary do not have phase names, hkl assignments or crystal-cell constraints.

5 Secondary geometry and reporting

Secondary search requires additional independent evidence, not merely a second similar basis. Invalid secondary fits are not presented as detected second patterns.

A substantial geometry-only second pattern can coexist with unresolved crystal phase identity. Conversely, two passing phase alternatives may explain the same peaks and need not establish two patterns.

6 Interpretation boundaries

Aligned HCP/BCC fits may be compatible with an orientation relationship, or may be alternative indexing of shared evidence. Pattern alignment alone cannot decide.

Projected fit, full 3D orientation, twinning, strain and chemical phase identity are different claims. Calibration, acquisition conditions and independent crystallographic checks remain necessary.

Grid orchestration, adaptive regions and batch runs

Each window calls the same detector and matcher. Window size changes the information available.

1 Uniform or adaptive grid

Uniform mode analyses the requested tessellation. Adaptive mode subdivides eligible unresolved/ambiguous or multiple-pattern regions, subject to minimum size.

Parent and child results are recorded separately; only active leaves contribute to the current map and region-area summaries.

2 Probe passing parents

Optional checks analyse four children even after a parent passes. A changed phase/zone, separate pattern, or mixed passing/unresolved children can justify a split.

All four children are published together. If every child merely loses usable evidence, keep the parent. This is a spatial heuristic, not a proof of homogeneity.

3 Advisory window-size check

At five paired sites, test 5%, 7.5%, 10% and 15% of the shorter image side using current manual/Auto settings.

Compare passing phase, zone and rotation across adjacent sizes. Stable ranges offer localization/resolution choices; supported dual geometry can favour smaller windows. The recommendation does not run a map automatically.

4 Smoothing and crop size

Fixed-bins mode retains σ bins. Fixed-nm^{-1} mode changes σ bins to preserve the selected reciprocal smoothing width.

The size check is not a smoothing optimizer. Too-small crops broaden or merge peaks; large crops may mix structures. The appropriate size need not be constant across an image.

5 Batch execution

Review/filter images and folders, capture a common settings snapshot and run grids with progress/estimated completion. Percentage windows resolve to a physical size per source.

Export each grid, optional display PNGs, and per-image/by-sample summaries. Enabled phases determine available columns. Failed/skipped images remain distinguishable from negative results.

6 Reweight and reproduce

Cached candidate evidence can be rescored after penalty changes. Existing windows and matches remain fixed; this does not search candidates omitted from the cache.

Detector, scale-search or sampling changes require fresh analysis. Current increasing-pass runs restart rather than extend the old job; incremental reuse remains proposed.

Overlapping windows and placement stability

Square FFT windows on square or triangular centre lattices; fitting precedes spatial aggregation.

1 Sampling layout

Hex centres: basis b1=(D,0), b2=(D/2,√3 D/2). Shifted passes use fixed fractions of these basis vectors, with pixel-rounded full windows.

Combined bounds are centred to balance margins. Window width, centre spacing and pass count are separate controls. Edges have fewer contributing samples; no cropped partial windows are fabricated.

2 Independent measurements

For each centre, extract the square physical window and run peak detection, Auto selection if enabled, template fitting and classification.

There is currently no shared spectral factorization or neighbour-assisted peak detection. Raw-centre symbols show individual fits; they are not the aggregated hex result.

3 Within-pass spatial weights

At map point x, $a_i = \exp\{-8[(\Delta x/W)^2 + (\Delta y/H)^2]\}$ within window i; $a_i=0$ outside.

For phase p and pass k: $f_{pk} = \Sigma_i a_i I(p \text{ passes in } i) / \Sigma_i a_i$. Missing phase fits give zero passing support. Observations in overlapping windows are correlated.

4 Equal-pass aggregation

F_p = mean_k f_{pk} across passes covering x. Placement disagreement = max_k f_{pk} − min_k f_{pk}; fewer than two passes gives zero.

Mean rank is computed only where a phase candidate exists, selecting its best passing fit or otherwise best rejected fit. Fit presence is reported separately. None of these fields is a confidence interval.

5 Display and separate structures

Raster and hex views evaluate the same weighted readings. Averaged hex winner = highest available mean rank. Passing support can blend phase colours; each phase frequency remains independent.

Separate-structure support counts windows explicitly classified as coexisting candidates. Two colours at one map location can instead come from different neighbouring windows.

6 Resolution and interpretation

Hex display spacing follows sampling spacing with a bounded tile count. Display tessellation does not make the square FFT footprints hexagonal or improve physical resolution.

Current support maps are not phase fractions. Proposed PCA/NMF would share Fourier evidence before matching, unlike this post-fit averaging; it requires separate validation.

Execution, persistence and auditability

Parallel work is isolated from display choices; saved records retain settings and source identity.

1 Parallel ROI analysis

Spawned processes analyse whole ROIs. Source pixels are mapped read-only from a temporary array file, avoiding repeated image transfer.

Numeric-library threads and worker FFT threads are limited to prevent nested oversubscription. Ordered publication preserves adaptive decisions. Worker failure and cancellation are surfaced to the UI.

2 Interactive responsiveness

Interactive analysis uses request-local cache containers, then merges bounded reusable entries. It does not hold the shared engine lock during expensive fitting.

Browser aborts/replaces obsolete requests and ignores stale results. An old numerical calculation may still finish in the background; switching source does not guarantee hard cancellation.

3 Caches and direct-mode threads

Prepared diffraction fields and detections can be reused when relevant source/settings match. Independent smoothing scales use a selectable thread pool.

Prediction coordinates and weak-peak preparation are reused within Auto trials. Direct crystal matching remains sequential; more smoothing threads do not parallelize every stage.

4 Saved state and exports

Local preferences retain basic controls. Project entries store named ROIs, tags, source references and analysis settings, with guarded atomic updates.

JSON exports retain numerical results, settings and provenance; map images render results. Reanalysis requires the original source pixels and compatible calibration, not just a screenshot.

5 Auxiliary acquisition map

Stage metadata provides approximate relative footprints, with per-reference coordinate rotation and remembered manual adjustments.

Nested reference navigation and crop previews assist inspection. This is not image registration or a verified stage calibration; it does not change the measured reciprocal vectors.

6 Reproducibility and scope

Record source identity, calibration, detector/search settings, manual edits, engine revision and runtime versions. Compare numerical tolerances as well as discrete labels across platforms.

Current implementation and future proposals are separated in the maintained plan. This atlas is reviewed on request; source hashes flag intervening code changes without automatically certifying correctness.